Supplement: Partial ports and crossing fluctuations in the three-dimensional Lorentz mirror model

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1 Microscopic geometry and exact identities

The cube has L^3 vertices, open faces in the transport direction and periodic transverse coordinates. The wide slab has dimensions $L \times 2L \times 2L$. The gapped right terminals occupy $y/L \in [1/8, 3/8)$ and $[5/8, 7/8)$. Wall A pairs adjacent unused ports along z using parity pairing; wall B pairs them along y, with the origin aligned to $L/8$. These are different microscopic wall rules, not samples to pool. The constriction restricts y, z to the central half-width in the central half of the transport direction; tangential pairing closes its sides. The retroreflecting variant returns a ray on its own incoming edge on the unused right ports. The sampling implementation specifies the discrete boundary conventions exactly.

In a fully open graph every connected component is a closed loop or an open path joining two boundary ports. Replacing a port in W by pure retroreflection reverses its path. An L–A component stays a crossing; an L–W component returns to its original left port; components without a left port cannot create a left-to-A crossing. Thus

$$C_{\mathrm{retro}} = n_{LA}^{\mathrm{open}}, \qquad f_L = n_{LW}^{\mathrm{open}}, \qquad N = C_{\mathrm{retro}} + f_L$$

for each environment, without independence or continuum assumptions. Ordinary tangential wall pairings can concatenate distinct paths and do not obey this proof. Same-terminal returns use two ports, whereas a pure retroreflection fixed point uses one. This distinction is retained in all counts; it was specifically corrected and independently checked before the original confirmation data.

For separated terminals write $n_1 = n_{LR_1}$, $n_2 = n_{LR_2}$, $n_3 = n_{R_1R_2}$, $\mu_i = \mathbb{E}n_i$, and $C_{ij} = \mathrm{Cov}(n_i, n_j)$. The original normalized identity residuals were

$$d_A = (C_{12} + C_{33} - 2\mu_3/3)/\mu_3, \tag{1}$$
$$d_B = (C_{13} + C_{33} - 2\mu_3/3)/\mu_3, \tag{2}$$
$$d_C = (C_{11} - C_{33} - 2(\mu_1 - \mu_3)/3)/\mu_1, \tag{3}$$
$$e_{12,13} = (C_{12} - C_{13})/\mu_3, \quad e_{12,23} = (C_{12} - C_{23})/\mu_3, \quad e_{13,23} = (C_{13} - C_{23})/\mu_3. \tag{4}$$

They are hypotheses about the smooth theory, not graph identities. Adjacent-terminal pilots were excluded from these tests by design because of contact contributions; they were not removed after a confirmation result.

2 Relation to established counting theory

2.1 Finite graph source and Gram matrices

The finite-graph connection polynomial is a useful exact representation: each open path contributes the scalar product of its terminal vectors, and closed loops have weight one in the physical replica limit. One may work with the polynomial before any analytic continuation. It does not establish a smooth phase, a convergent replica continuation or an interchange of thermodynamic and source-derivative limits.

For the multiterminal conductor representation of Nazarov–Bagrets [1], take

$$G_L = \begin{pmatrix} -1 & -2 \\ 0 & 1 \end{pmatrix}, \quad G_A = \begin{pmatrix} 1 & 0 \\ -2q & -1 \end{pmatrix}, \quad G_W = \begin{pmatrix} 1 & 0 \\ -2 & -1 \end{pmatrix}.$$

Each matrix squares to the identity. The L, A bilinear product $\frac{1}{2}\operatorname{Tr} G_L G_A = 2q-1$; the other different-terminal products equal one. Consequently $q = (1+z)/2$ gives the source counting only L, A paths. The Pauli-vector identity turns the specified leading gradient action into the same complex-sphere stationary problem, up to normalization. The transformation is singular at zero source in this coordinate chart; the correspondence is understood as an analytic generating-function expansion near that point. It is not an equality of microscopic measures or of all fluctuation corrections.

With conductor mean twice the crossing mean, the composite generating function gives

$$\kappa_{1,C} = \kappa_{1,e}/2, \quad \kappa_{2,C} = (\kappa_{1,e} + \kappa_{2,e})/4, \quad \kappa_{3,C} = (3\kappa_{2,e} + \kappa_{3,e})/8.$$

For characteristic potentials f, g, harmonic integration by parts gives

$$\int f|\nabla g|^2 = M/2, \qquad \int f^2|\nabla g|^2 = \int |\nabla(fg)|^2 = J.$$

The established noise functional [2, 3] therefore gives $v_e = 1-2J/M$, followed by $v_C = 1-J/M$. At an adjacent counted/uncounted contact, retain a smooth regulator before taking its limit; the left potential vanishes there. These general structures are inherited, not claimed as a new framework.

2.2 Mixed twist

For disjoint right terminals with harmonic potentials ϕ, ψ, use boundary spins $(1,0,0)$, $(\cos a, \sin a, 0)$, $(\cos b, 0, \sin b)$ and action $K\int |\nabla \mathbf{n}|^2/2$. In transverse coordinates $\mathbf{n} = (\sqrt{1-\pi^2}, \pi_1, \pi_2)$, stationarity removes the contribution of mixed field corrections at order a^2b^2. Integration by parts yields

$$I = \int \phi\psi \nabla\phi \cdot \nabla\psi = -J/2, \qquad [a^2b^2]S_{\min} = KI = -KJ/2.$$

Independent angular coordinates give $-K\int \psi^2|\nabla\phi|^2/2$, the same coefficient. Since the connection generator is $\mathbb{E}[(\cos a)^X(\cos b)^Y]$, $X = n_1 + n_3, Y = n_2 + n_3$, its mixed logarithmic coefficient is $\mathrm{Cov}(X,Y)/4$. Thus $\mathrm{Cov}(X,Y)/\mu_3 = 2J/M$. This assumes a smooth stationary branch and that each microscopic side wall has the imposed Neumann limit.

The registered geometric target is 0.5138587342, with deterministic allowance 0.0018719596. It was obtained from two boundary discretizations, two discrete functionals and both extrapolation forms, retaining all grids from 16 to 512. This allowance covers those numerical differences, not microscopic finite-size effects or a rigorous continuum error. It is subtracted before forming standardized discrepancies, not combined as an independent Gaussian error.

2.3 Stripe and higher-source formulas

On the unit cube, the harmonic mode is $\sinh(kx)/\sinh k$. Its squared integral is $I(k) = \coth k/(2k) - 1/(2\sinh^2 k)$. For rational area fractions, the inverse-cubic Fourier sum may be evaluated independently by grouping residue classes and using Hurwitz zeta functions. Direct quadrature of the mode integral gives a second check of the kernel. The public script repeats both checks for all tabulated stripe targets. For the quarter-open square, a two-dimensional sum at cutoff 8192 gives

$$0.8721154649612308 \le v_{\square} \le 0.8721154659224014.$$

This bounds the continuum Fourier tail only. Frozen statistical tests keep their original registered targets, even where the later deterministic calculation supplies more digits.

For a transverse Fourier wave number k, the conditional sixth-order source calculation gives the mixed third-cumulant kernel

$$T_*(k) = \frac{2}{3} - \frac{2\coth k}{3k} + \frac{2}{3\sinh^2 k} - \frac{\coth k}{2k\sinh^2 k} + \frac{1}{4k^2\sinh^2 k} + \frac{\coth k}{4k^3}. \tag{5}$$

It tends to $4/15$ at zero wave number. For a translation-averaged mask with $C_A = pN + D_A$ and $Q_j = \langle D_A^j \rangle_{\text{translations}}$,

$$\kappa_3(C_A) = p^3\kappa_3(N) + 3p\,\mathrm{Cov}(N, Q_2) + \mathbb{E}Q_3.$$

For the half-open equal-width stripes the translated complement implies $Q_3 = 0$ separately in each environment. Hence the source projection is

$$h_A = p^2 T_*(0) + 3\sum_{k\neq 0} |a_k|^2 T_*(k), \qquad p = 1/2.$$

This yields the refined one-stripe target 0.4994424165286316. Quarter-open masks generally have nonzero Q_3; that term cannot be dropped. A sixth-order source refers here to derivatives of the terminal source, not to the sixth count cumulant. These calculations do not complete the microscopic matching of higher-gradient boundary terms.

3 Statistical rules and complete original results

The original study collected 10,446,000 environments, then a fixed addition of 18,167,000 for the prescribed higher-cumulant and retroreflection tests. The latter did not replace the original second-cumulant and covariance samples. All reported uncertainties below are bootstrap standard errors unless explicitly identified as deterministic allowances. Unbiased sample cumulants are used before forming plug-in ratios; a ratio is not asserted unbiased. Each environment, not each path or translated mask, is an independent resampling unit.

The fixed forms are $a + b/L$ and $a + b/L + c/L^2$. Weighted least squares uses inverse squared empirical bootstrap errors; each resample is refitted. All original fits use $L = 16, 32, 64, 96$. Fit probabilities below 0.01 prevent a decision. A rejection requires same-sign discrepancies exceeding three standard errors in both rules and both forms, the trend guard and Holm significance. The trend guard prevents rejecting a target when the largest-size point is clearly moving toward it. The per-test probability is the maximum over the four rule/form probabilities. Holm uses all 30 tests at $\alpha = 0.002699796063260207$, including secondary and inadequately powered observables. Fourth-cumulant uncertainty also has a registered 40-block stability check. Failure of precision never becomes evidence for agreement or rejection.

The original two-dimensional baseline passed only a limited-precision checkpoint after finite-size correction. It did not reproduce the cited baseline at the original publication's precision. This remains an explicit limitation; passing that checkpoint cannot validate all higher cumulants.

These plans were recorded locally before collection, without an independent timestamp service. Hashes authenticate contents, not the historical date. The public registration files contain the exact precision caps, guards, seed allocations and original machine-readable labels.

Table 1: All 30 original decisions. C: consistency at attained precision; F: fit criterion failed; P: insufficient precision; D: disagreement or guard; R: robust rejection. C is not an equivalence result.

ID	Geometry	Observable	Target	Result	p
1	Cube	v	0.666667	C	0.984
2	Wide slab	v	0.666667	C	0.978

ID	Geometry	Observable	Target	Result	p
3	Gap, wall A	v	0.666667	C	0.981
4	Gap, wall B	v	0.666667	C	0.987
5	Constriction	v	0.666667	F	0.971
6	Gap, wall A	d_A	0.000000	C	0.358
7	Gap, wall A	d_B	0.000000	D	0.482
8	Gap, wall A	d_C	0.000000	C	0.411
9	Gap, wall A	$e_{12,13}$	0.000000	C	0.775
10	Gap, wall A	$e_{12,23}$	0.000000	D	0.41
11	Gap, wall A	$e_{13,23}$	0.000000	C	0.976
12	Gap, wall B	d_A	0.000000	C	0.313
13	Gap, wall B	d_B	0.000000	D	0.579
14	Gap, wall B	d_C	0.000000	C	0.965
15	Gap, wall B	$e_{12,13}$	0.000000	D	0.791
16	Gap, wall B	$e_{12,23}$	0.000000	D	0.58
17	Gap, wall B	$e_{13,23}$	0.000000	C	0.728
18	Cube	h	0.266667	C	0.674
19	Cube	κ_4/κ_1	-0.076190	P	0.99
20	Wide slab	h	0.266667	P	0.756
21	Wide slab	κ_4/κ_1	-0.076190	P	0.673
22	Gap, wall A	h	0.266667	C	0.448
23	Gap, wall A	κ_4/κ_1	-0.076190	P	0.773
24	Gap, wall B	h	0.266667	F	0.744
25	Gap, wall B	κ_4/κ_1	-0.076190	P	0.768
26	Constriction	h	0.266667	C	0.884
27	Constriction	κ_4/κ_1	-0.076190	P	0.894
28	Gap, wall A	$\mathrm{Cov}(X,Y)/\mu_3$	0.513859	C	1
29	Gap, wall B	$\mathrm{Cov}(X,Y)/\mu_3$	0.513859	C	0.613
30	Retroreflection	v	0.666667	R	7.02e-223

Table 2: Original Holm step-down sequence across all 30 tests; the family level is the fixed two-sided three-standard-error probability. Rejection also requires the physical and fit guards.

Rank	ID	p	Threshold	Step rejects
1	30	7.02e-223	9e-05	yes
2	12	0.313	9.31e-05	no
3	6	0.358	9.64e-05	no
4	10	0.41	0.0001	no
5	8	0.411	0.000104	no
6	22	0.448	0.000108	no
7	7	0.482	0.000112	no
8	13	0.579	0.000117	no
9	16	0.58	0.000123	no
10	29	0.613	0.000129	no
11	21	0.673	0.000135	no
12	18	0.674	0.000142	no
13	17	0.728	0.00015	no
14	24	0.744	0.000159	no
15	20	0.756	0.000169	no
16	25	0.768	0.00018	no
17	23	0.773	0.000193	no
18	9	0.775	0.000208	no
19	15	0.791	0.000225	no
20	26	0.884	0.000245	no
21	27	0.894	0.00027	no
22	14	0.965	0.0003	no
23	5	0.971	0.000337	no
24	11	0.976	0.000386	no
25	2	0.978	0.00045	no
26	3	0.981	0.00054	no
27	1	0.984	0.000675	no

Rank	ID	p	Threshold	Step rejects
28	4	0.987	0.0009	no
29	19	0.99	0.00135	no
30	28	1	0.0027	no

Table 3: All 120 original fits. S/O: standard/orthogonal; degree 1/2 means the prescribed linear/quadratic fit in inverse size. All use sizes 16, 32, 64, 96.

ID	Rule	Degree	Intercept	p_{fit}	χ^2	dof
1	S	1	0.670186 ± 0.002851	0.536	1.247	2
1	S	2	0.666812 ± 0.007241	0.324	0.971	1
1	O	1	0.669290 ± 0.003060	0.979	0.043	2
1	O	2	0.669803 ± 0.006917	0.849	0.036	1
2	S	1	0.666763 ± 0.003554	0.584	1.074	2
2	S	2	0.670186 ± 0.007700	0.365	0.822	1
2	O	1	0.668777 ± 0.003655	0.139	3.951	2
2	O	2	0.679541 ± 0.008139	0.186	1.747	1
3	S	1	0.666694 ± 0.001133	0.834	0.363	2
3	S	2	0.665997 ± 0.002441	0.611	0.259	1
3	O	1	0.667508 ± 0.001084	0.95	0.103	2
3	O	2	0.666846 ± 0.002343	0.965	0.002	1
4	S	1	0.667469 ± 0.001125	0.281	2.541	2
4	S	2	0.664941 ± 0.002457	0.27	1.218	1
4	O	1	0.667941 ± 0.001065	0.28	2.545	2
4	O	2	0.666627 ± 0.002409	0.141	2.163	1
5	S	1	0.664228 ± 0.003253	0.856	0.312	2
5	S	2	0.666938 ± 0.007454	0.698	0.150	1
5	O	1	0.666232 ± 0.003379	0.0133	8.643	2
5	O	2	0.672217 ± 0.007605	0.00503	7.869	1
6	S	1	0.004208 ± 0.001444	0.387	1.896	2
6	S	2	0.002873 ± 0.003124	0.197	1.666	1
6	O	1	0.002331 ± 0.001406	0.569	1.126	2
6	O	2	0.004033 ± 0.002975	0.394	0.726	1
7	S	1	0.007607 ± 0.001373	0.0643	5.487	2
7	S	2	0.002169 ± 0.002919	0.281	1.162	1
7	O	1	0.003383 ± 0.001317	0.805	0.434	2
7	O	2	0.001999 ± 0.002839	0.709	0.139	1
8	S	1	-0.003859 ± 0.002189	0.248	2.788	2
8	S	2	-0.004396 ± 0.004642	0.096	2.772	1
8	O	1	-0.001830 ± 0.002224	0.327	2.238	2
8	O	2	-0.007915 ± 0.004752	0.651	0.204	1
9	S	1	-0.003226 ± 0.001081	0.166	3.595	2
9	S	2	0.000678 ± 0.002369	0.711	0.137	1
9	O	1	-0.000976 ± 0.001011	0.262	2.677	2
9	O	2	0.001940 ± 0.002208	0.499	0.458	1
10	S	1	-0.003536 ± 0.001122	0.0248	7.397	2
10	S	2	0.002017 ± 0.002448	0.451	0.569	1
10	O	1	-0.001053 ± 0.001003	0.262	2.677	2
10	O	2	0.002171 ± 0.002264	0.894	0.018	1
11	S	1	-0.000322 ± 0.001219	0.727	0.638	2
11	S	2	0.001330 ± 0.002651	0.722	0.127	1
11	O	1	-0.000034 ± 0.001099	0.759	0.552	2
11	O	2	0.000226 ± 0.002420	0.464	0.537	1
12	S	1	0.002980 ± 0.001397	0.948	0.106	2
12	S	2	0.003248 ± 0.003219	0.755	0.097	1
12	O	1	0.001887 ± 0.001394	0.753	0.568	2
12	O	2	0.003539 ± 0.003024	0.665	0.188	1
13	S	1	0.007519 ± 0.001267	0.13	4.086	2
13	S	2	0.002491 ± 0.002968	0.586	0.297	1
13	O	1	0.003508 ± 0.001297	0.652	0.857	2
13	O	2	0.001568 ± 0.002829	0.617	0.251	1

ID	Rule	Degree	Intercept	p_{fit}	χ^2	dof
14	S	1	-0.000777 ± 0.002097	0.534	1.256	2
14	S	2	-0.005089 ± 0.004825	0.633	0.228	1
14	O	1	0.000100 ± 0.002256	0.401	1.828	2
14	O	2	-0.005746 ± 0.004972	0.884	0.021	1
15	S	1	-0.004280 ± 0.001091	0.0328	6.837	2
15	S	2	0.000614 ± 0.002311	0.26	1.268	1
15	O	1	-0.001566 ± 0.000972	0.195	3.266	2
15	O	2	0.001961 ± 0.002239	0.958	0.003	1
16	S	1	-0.003668 ± 0.001069	0.826	0.382	2
16	S	2	-0.002941 ± 0.002316	0.61	0.261	1
16	O	1	-0.001088 ± 0.000967	0.112	4.382	2
16	O	2	0.001198 ± 0.002165	0.0845	2.976	1
17	S	1	0.000726 ± 0.001168	0.139	3.946	2
17	S	2	-0.003574 ± 0.002510	0.58	0.307	1
17	O	1	0.000371 ± 0.001067	0.236	2.886	2
17	O	2	-0.000911 ± 0.002334	0.113	2.509	1
18	S	1	0.262196 ± 0.010621	0.583	1.078	2
18	S	2	0.251233 ± 0.029269	0.337	0.920	1
18	O	1	0.244814 ± 0.009939	0.125	4.161	2
18	O	2	0.193727 ± 0.026560	0.875	0.025	1
19	S	1	-0.045590 ± 0.137130	0.172	3.525	2
19	S	2	-0.081126 ± 0.407419	0.0608	3.516	1
19	O	1	-0.049209 ± 0.108557	0.583	1.080	2
19	O	2	-0.353606 ± 0.331199	0.719	0.129	1
20	S	1	0.202972 ± 0.093313	0.276	2.573	2
20	S	2	0.027818 ± 0.218077	0.187	1.744	1
20	O	1	0.238415 ± 0.079523	0.846	0.334	2
20	O	2	0.326967 ± 0.194047	0.812	0.057	1
21	S	1	-3.979270 ± 2.546258	0.826	0.383	2
21	S	2	-6.208214 ± 7.108450	0.602	0.272	1
21	O	1	-0.875423 ± 1.891749	0.136	3.997	2
21	O	2	-2.837248 ± 5.039755	0.0506	3.820	1
22	S	1	0.242432 ± 0.009140	0.599	1.025	2
22	S	2	0.233002 ± 0.022319	0.367	0.815	1
22	O	1	0.273540 ± 0.009065	0.0411	6.382	2
22	O	2	0.300637 ± 0.021756	0.0301	4.701	1
23	S	1	-0.110891 ± 0.120292	0.0992	4.622	2
23	S	2	-0.333729 ± 0.331838	0.0425	4.117	1
23	O	1	-0.021106 ± 0.100386	0.23	2.938	2
23	O	2	-0.252580 ± 0.269304	0.15	2.075	1
24	S	1	0.263511 ± 0.009659	0.00709	9.898	2
24	S	2	0.198954 ± 0.023586	0.327	0.962	1
24	O	1	0.272525 ± 0.008741	0.531	1.267	2
24	O	2	0.253883 ± 0.022015	0.514	0.425	1
25	S	1	-0.145524 ± 0.123678	0.351	2.092	2
25	S	2	-0.458751 ± 0.350262	0.284	1.148	1
25	O	1	0.009425 ± 0.102289	0.775	0.510	2
25	O	2	-0.158496 ± 0.279459	0.778	0.080	1
26	S	1	0.265247 ± 0.009746	0.432	1.681	2
26	S	2	0.241190 ± 0.025746	0.394	0.726	1
26	O	1	0.272589 ± 0.010409	0.791	0.470	2
26	O	2	0.280407 ± 0.027665	0.539	0.377	1
27	S	1	-0.149707 ± 0.076068	0.665	0.815	2
27	S	2	-0.105123 ± 0.217362	0.381	0.767	1
27	O	1	-0.012357 ± 0.070089	0.449	1.603	2
27	O	2	-0.204942 ± 0.199979	0.451	0.569	1
28	S	1	0.517004 ± 0.001513	0.522	1.301	2
28	S	2	0.514276 ± 0.003236	0.533	0.389	1
28	O	1	0.514532 ± 0.001441	0.886	0.242	2
28	O	2	0.514961 ± 0.003056	0.641	0.218	1
29	S	1	0.517263 ± 0.001422	0.995	0.010	2
29	S	2	0.517552 ± 0.003278	0.999	0.000	1
29	O	1	0.517928 ± 0.001388	0.955	0.092	2
29	O	2	0.517246 ± 0.002997	0.871	0.027	1

ID	Rule	Degree	Intercept	p_{fit}	χ^2	dof
30	S	1	0.822143 ± 0.002181	0.116	4.307	2
30	S	2	0.822773 ± 0.004819	0.0384	4.286	1
30	O	1	0.818848 ± 0.002116	0.173	3.509	2
30	O	2	0.813176 ± 0.004597	0.203	1.618	1

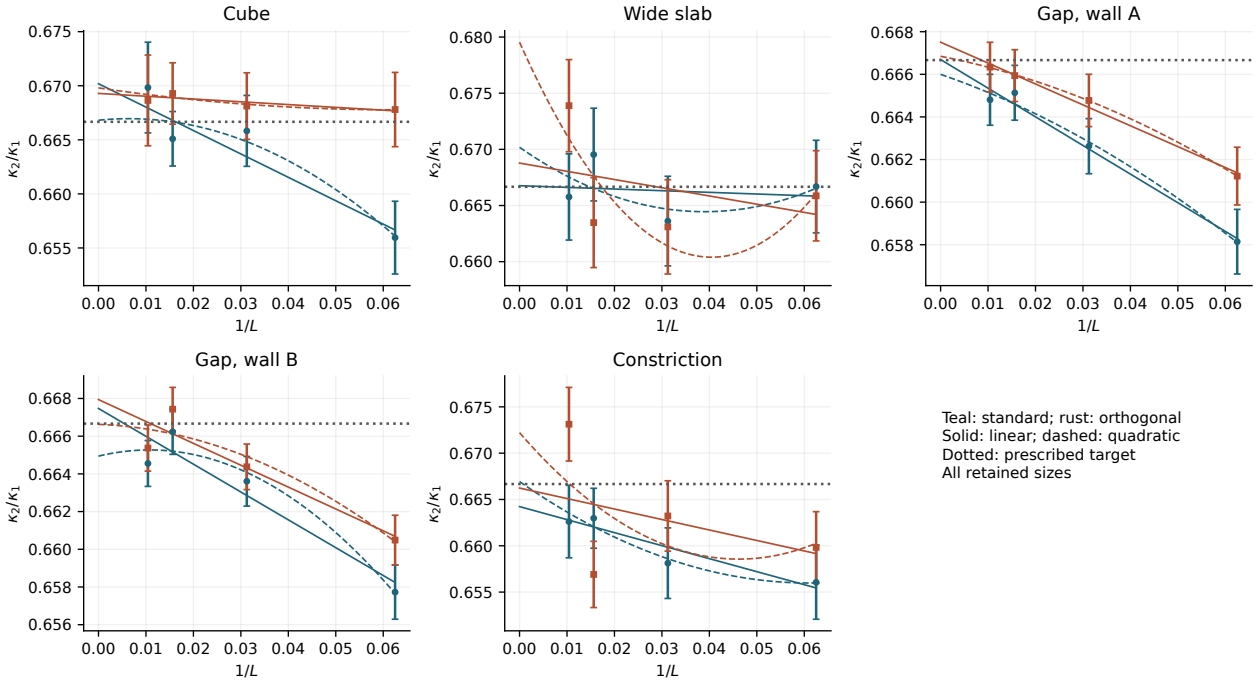


Figure 1: Original second-cumulant ratios. All original sizes, both rules and both prescribed forms are shown, including the constriction's failed fit criterion.

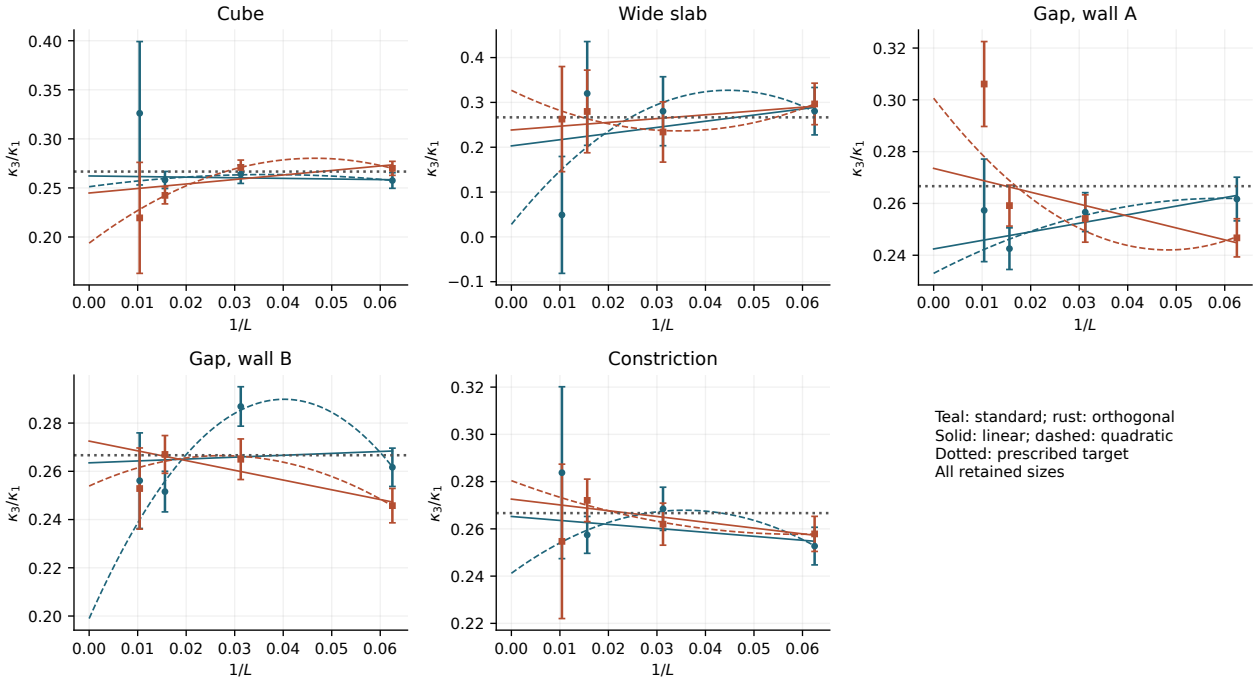


Figure 2: Original third-cumulant ratios after the fixed addition where planned. The wide slab remains underpowered, and wall B does not pass the fit criterion.

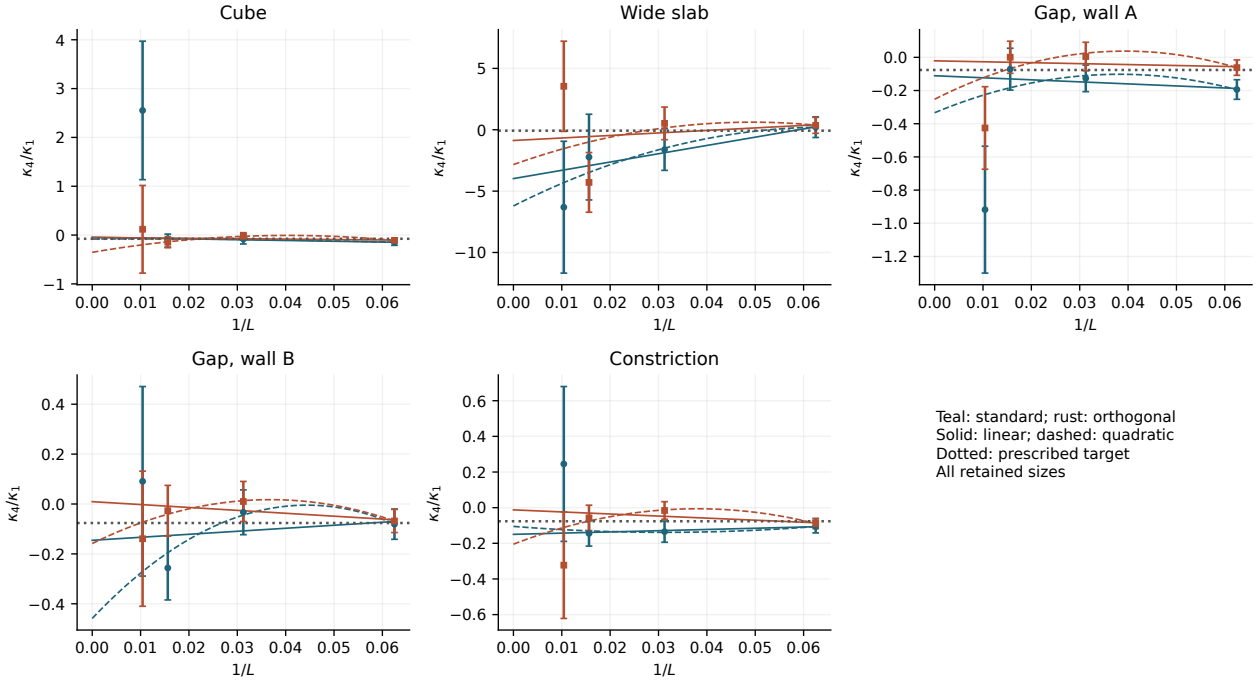


Figure 3: Original fourth-cumulant ratios. No geometry supplies a conclusive fourth-cumulant decision.

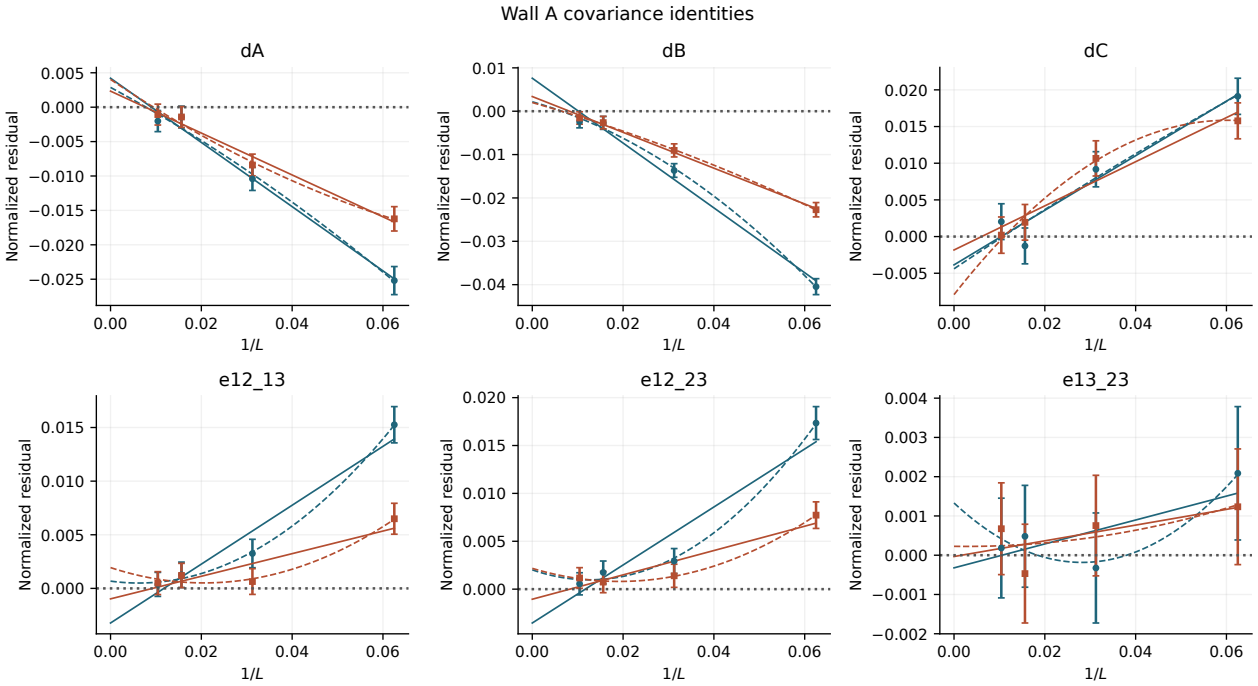


Figure 4: All six gapped-terminal identity residuals for wall A. Solid/dashed curves are linear/quadratic fits; teal/rust denote standard/orthogonal rules.

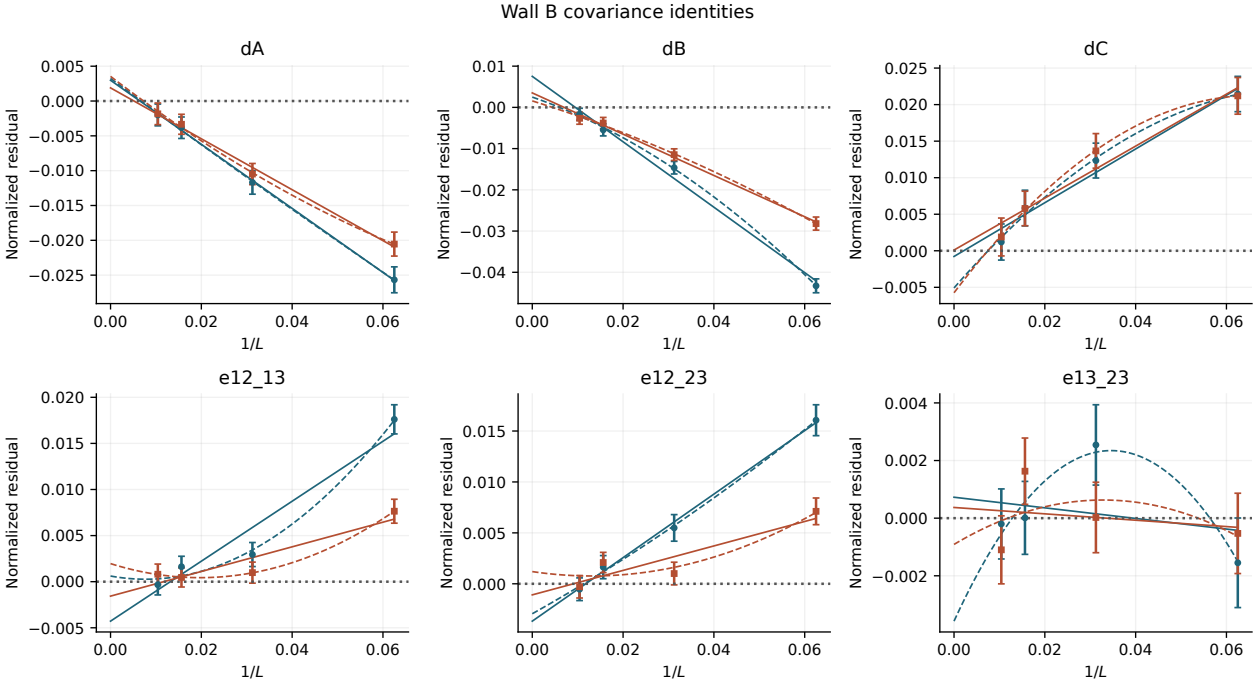


Figure 5: The corresponding six identity residuals for wall B. Together the two walls leave five identity tests undecided.

Table 4: Complete original sample allocation. The fixed addition does not change the original second-cumulant or covariance samples.

Data	Geometry	Rule	L	Environments
Initial	Cube	O	16	74,000
Initial	Cube	O	32	92,000
Initial	Cube	O	64	108,000
Initial	Cube	O	96	54,000
Initial	Cube	S	16	72,000
Initial	Cube	S	32	83,000
Initial	Cube	S	64	142,000
Initial	Cube	S	96	55,000
Initial	Gap, wall A	O	16	473,000
Initial	Gap, wall A	O	32	578,000
Initial	Gap, wall A	O	64	603,000
Initial	Gap, wall A	O	96	620,000
Initial	Gap, wall A	S	16	391,000
Initial	Gap, wall A	S	32	515,000
Initial	Gap, wall A	S	64	555,000
Initial	Gap, wall A	S	96	575,000
Initial	Constriction	O	16	57,000
Initial	Constriction	O	32	58,000
Initial	Constriction	O	64	65,000
Initial	Constriction	O	96	55,000
Initial	Constriction	S	16	57,000
Initial	Constriction	S	32	58,000
Initial	Constriction	S	64	86,000
Initial	Constriction	S	96	58,000
Initial	Wide slab	O	16	54,000
Initial	Wide slab	O	32	52,000
Initial	Wide slab	O	64	53,000
Initial	Wide slab	O	96	53,000
Initial	Wide slab	S	16	52,000
Initial	Wide slab	S	32	54,000
Initial	Wide slab	S	64	52,000
Initial	Wide slab	S	96	57,000
Initial	Gap, wall B	O	16	490,000
Initial	Gap, wall B	O	32	615,000
Initial	Gap, wall B	O	64	673,000
Initial	Gap, wall B	O	96	604,000
Initial	Gap, wall B	S	16	415,000
Initial	Gap, wall B	S	32	535,000
Initial	Gap, wall B	S	64	603,000
Initial	Gap, wall B	S	96	600,000
Fixed addition	Retroreflection	O	16	218,000
Fixed addition	Retroreflection	O	32	237,000
Fixed addition	Retroreflection	O	64	230,000
Fixed addition	Retroreflection	O	96	238,000
Fixed addition	Retroreflection	S	16	212,000
Fixed addition	Retroreflection	S	32	217,000
Fixed addition	Retroreflection	S	64	220,000
Fixed addition	Retroreflection	S	96	210,000
Fixed addition	Cube	O	16	458,000
Fixed addition	Cube	O	32	921,000
Fixed addition	Cube	O	64	1,462,000
Fixed addition	Cube	S	16	517,000
Fixed addition	Cube	S	32	885,000
Fixed addition	Cube	S	64	2,253,000
Fixed addition	Gap, wall A	O	32	56,000
Fixed addition	Gap, wall A	O	64	1,231,000
Fixed addition	Gap, wall A	S	16	103,000
Fixed addition	Gap, wall A	S	32	846,000
Fixed addition	Gap, wall A	S	64	1,938,000
Fixed addition	Constriction	O	16	110,000
Fixed addition	Constriction	O	32	203,000

Data	Geometry	Rule	L	Environments
Fixed addition	Constriction	O	64	426,000
Fixed addition	Constriction	S	16	135,000
Fixed addition	Constriction	S	32	261,000
Fixed addition	Constriction	S	64	826,000
Fixed addition	Gap, wall B	O	32	155,000
Fixed addition	Gap, wall B	O	64	1,164,000
Fixed addition	Gap, wall B	S	16	117,000
Fixed addition	Gap, wall B	S	32	626,000
Fixed addition	Gap, wall B	S	64	1,692,000

4 Stripe confirmation: complete reports

The layout-difference family has four tests: zero difference and the predicted difference, each with both forms. The family level is 0.05, the fit threshold 0.01, the rule-agreement bound three combined standard errors and the maximum joint standard error 0.0075. The later absolute-ratio family has six tests: three masks, each with both forms. Its corresponding precision bound is 0.003. Each form has a common intercept with separate rule-dependent corrections. Every separate-rule fit is also reported.

Raw fixed-mask ratios, the full-face ratio, the layout difference within the absolute-data sample, higher cumulants and spectra are auxiliary where not listed in the respective primary plan. In particular, the auxiliary third-cumulant and spectrum diagnostics were registered during collection, not before it, and receive no new confirmation decision. The compact release stores every environment in both primary stripe samples. For the later sample its columns are $(N, 2LQ_1, 2LQ_2, 2LQ_4, C_1, C_2, C_4)$, all exact integers. This preserves the joint empirical distribution used by the primary bootstrap. Full transverse marginals are needed for the auxiliary higher-order and spectrum replay and are not included in these compact columns. They can be regenerated from the original seeds.

Table 5: Stripe sample allocation and fixed bootstrap seeds. Physical seeds are recorded separately.

Data	Rule	L	Environments	Bootstrap seed
Layout	Standard	16	293,000	330030001
Layout	Standard	32	616,500	330030002
Layout	Standard	64	251,750	330030003
Layout	Standard	128	20,000	330030004
Layout	Orthogonal	16	327,500	330030005
Layout	Orthogonal	32	698,750	330030006
Layout	Orthogonal	64	294,750	330030007
Layout	Orthogonal	128	20,000	330030008
Absolute	Standard	32	166,000	679925300
Absolute	Standard	64	290,250	841496916
Absolute	Standard	128	77,000	4285283429
Absolute	Standard	256	40,000	1600146856
Absolute	Orthogonal	32	166,000	4258797266
Absolute	Orthogonal	64	290,250	2141657993
Absolute	Orthogonal	128	77,000	3227410029
Absolute	Orthogonal	256	40,000	1894540754

Table 6: All raw primary probabilities and Holm rejections for the two independent stripe families.

Family	Test	p	Rejected
Layout	zero_1	7.23e-37	yes
Layout	theory_1	0.543	no
Layout	zero_2	2.33e-06	yes
Layout	theory_2	0.957	no
Absolute	r1_1	0.762	no
Absolute	r1_2	0.319	no
Absolute	r2_1	0.528	no

Family	Test	p	Rejected
Absolute	r2_2	0.146	no
Absolute	r4_1	0.381	no
Absolute	r4_2	0.0589	no

Table 7: All absolute-data fits, including full-face, raw fixed-mask and contrast diagnostics. The three symmetry-averaged stripe ratios alone belong to the primary family.

Rule	Quantity	Degree	Intercept	p_{fit}
S	rN	1	0.665906 ± 0.002788	0.232
S	rN	2	0.663579 ± 0.005929	0.0987
S	r1	1	0.800636 ± 0.001598	0.605
S	r1	2	0.799705 ± 0.003417	0.341
S	r2	1	0.816910 ± 0.001632	0.072
S	r2	2	0.812558 ± 0.003494	0.0716
S	r4	1	0.826134 ± 0.001635	0.158
S	r4	2	0.821426 ± 0.003508	0.255
S	raw1	1	0.799957 ± 0.003422	0.831
S	raw1	2	0.798515 ± 0.007165	0.573
S	raw2	1	0.814736 ± 0.003422	0.123
S	raw2	2	0.801480 ± 0.007376	0.969
S	raw4	1	0.823500 ± 0.003396	0.608
S	raw4	2	0.817844 ± 0.007452	0.619
S	delta41	1	0.025500 ± 0.001181	0.205
S	delta41	2	0.021746 ± 0.002424	0.769
O	rN	1	0.665209 ± 0.002764	0.163
O	rN	2	0.655500 ± 0.005913	0.612
O	r1	1	0.798848 ± 0.001592	0.3
O	r1	2	0.794292 ± 0.003409	0.735
O	r2	1	0.817260 ± 0.001609	0.361
O	r2	2	0.813192 ± 0.003455	0.587
O	r4	1	0.825585 ± 0.001628	0.0772
O	r4	2	0.818978 ± 0.003458	0.472
O	raw1	1	0.798301 ± 0.003280	0.218
O	raw1	2	0.791874 ± 0.007167	0.155
O	raw2	1	0.819102 ± 0.003537	0.671
O	raw2	2	0.813257 ± 0.007355	0.929
O	raw4	1	0.825974 ± 0.003384	0.255
O	raw4	2	0.815976 ± 0.007290	0.548
O	delta41	1	0.026691 ± 0.001174	0.209
O	delta41	2	0.024545 ± 0.002521	0.139
J	rN	1	0.665560 ± 0.001945	0.253
J	rN	2	0.659580 ± 0.004139	0.272
J	r1	1	0.799752 ± 0.001116	0.545
J	r1	2	0.797022 ± 0.002399	0.514
J	r2	1	0.817086 ± 0.001130	0.198
J	r2	2	0.812874 ± 0.002408	0.314
J	r4	1	0.825861 ± 0.001150	0.114
J	r4	2	0.820215 ± 0.002456	0.559
J	raw1	1	0.799132 ± 0.002388	0.618
J	raw1	2	0.795211 ± 0.005087	0.429
J	raw2	1	0.816903 ± 0.002423	0.327
J	raw2	2	0.807288 ± 0.005145	0.733
J	raw4	1	0.824742 ± 0.002427	0.552
J	raw4	2	0.816904 ± 0.005223	0.887
J	delta41	1	0.026081 ± 0.000827	0.235
J	delta41	2	0.023103 ± 0.001750	0.405

Table 8: All layout-data fits. Absolute ratios are secondary diagnostics; the paired difference is primary.

Rule	Quantity	Degree	Intercept	p_{fit}
S	rN	1	0.668690 ± 0.002013	0.621
S	rN	2	0.671930 ± 0.005149	0.49
S	r1	1	0.807745 ± 0.002424	0.374
S	r1	2	0.802115 ± 0.005792	0.35
S	r2	1	0.822256 ± 0.002470	0.582
S	r2	2	0.822334 ± 0.006252	0.298
S	r4	1	0.831697 ± 0.002369	0.652
S	r4	2	0.826867 ± 0.006135	0.759
S	delta41	1	0.023946 ± 0.003126	0.89
S	delta41	2	0.024733 ± 0.007817	0.638
O	rN	1	0.664260 ± 0.001875	0.643
O	rN	2	0.668104 ± 0.004524	0.824
O	r1	1	0.799584 ± 0.002178	0.853
O	r1	2	0.796940 ± 0.005645	0.826
O	r2	1	0.818948 ± 0.002270	0.909
O	r2	2	0.820895 ± 0.005799	0.819
O	r4	1	0.828765 ± 0.002302	0.424
O	r4	2	0.822528 ± 0.005842	0.555
O	delta41	1	0.029144 ± 0.002854	0.711
O	delta41	2	0.025521 ± 0.007349	0.531
J	Δv	1	0.026719 ± 0.002106	0.791
J	Δv	2	0.025149 ± 0.005325	0.892

5 Retrospective comparisons and unfavorable diagnostics

The older standard-rule data show a common linear offset of $0.0070564322 \pm 0.0016158417$ in the three absolute stripe ratios. Their covariance-aware shape test after removing a common offset has $p = 0.7386$. This distinguishes a layout contrast from an absolute test; it does not justify removing the offset from confirmation data. All-size pooled fits below retain this upward linear tendency, including $L = 16$; the quadratic fits and fresh larger-size sample behave differently. Curvature is a possible explanation only.

Table 9: All retrospective all-size pooled fits. Fixed uses fixed-mask estimators in both batches; Mixed uses orbit averaging for the later batch. Neither changes a primary analysis.

Method	Rule	Quantity	Degree	Intercept	p_{fit}
Fixed	S	r1	1	0.805533 ± 0.001741	0.409
Fixed	S	r1	2	0.799387 ± 0.003393	0.902
Fixed	S	r2	1	0.820625 ± 0.001749	0.152
Fixed	S	r2	2	0.815686 ± 0.003489	0.245
Fixed	S	r4	1	0.829422 ± 0.001719	0.437
Fixed	S	r4	2	0.823008 ± 0.003361	0.937
Fixed	O	r1	1	0.799968 ± 0.001622	0.633
Fixed	O	r1	2	0.797655 ± 0.003299	0.599
Fixed	O	r2	1	0.819273 ± 0.001689	0.982
Fixed	O	r2	2	0.819692 ± 0.003402	0.956
Fixed	O	r4	1	0.827269 ± 0.001663	0.492
Fixed	O	r4	2	0.823076 ± 0.003328	0.64
Fixed	J	r1	1	0.802640 ± 0.001182	0.253
Fixed	J	r1	2	0.798505 ± 0.002382	0.91
Fixed	J	r2	1	0.819917 ± 0.001193	0.627
Fixed	J	r2	2	0.817740 ± 0.002393	0.673
Fixed	J	r4	1	0.828301 ± 0.001218	0.522
Fixed	J	r4	2	0.823042 ± 0.002337	0.946
Mixed	S	r1	1	0.803204 ± 0.001119	0.153
Mixed	S	r1	2	0.799128 ± 0.001948	0.747
Mixed	S	r2	1	0.818771 ± 0.001118	0.138
Mixed	S	r2	2	0.815967 ± 0.002024	0.243

Method	Rule	Quantity	Degree	Intercept	p_{fit}
Mixed	S	r4	1	0.828672 ± 0.001107	0.156
Mixed	S	r4	2	0.824458 ± 0.001983	0.765
Mixed	O	r1	1	0.800309 ± 0.001082	0.446
Mixed	O	r1	2	0.797970 ± 0.001906	0.606
Mixed	O	r2	1	0.818049 ± 0.001122	0.847
Mixed	O	r2	2	0.817015 ± 0.001923	0.811
Mixed	O	r4	1	0.827050 ± 0.001118	0.211
Mixed	O	r4	2	0.823700 ± 0.001930	0.529
Mixed	J	r1	1	0.801727 ± 0.000769	0.137
Mixed	J	r1	2	0.798547 ± 0.001343	0.839
Mixed	J	r2	1	0.818399 ± 0.000781	0.48
Mixed	J	r2	2	0.816501 ± 0.001370	0.61
Mixed	J	r4	1	0.827847 ± 0.000800	0.131
Mixed	J	r4	2	0.824077 ± 0.001366	0.816

Table 10: All retrospective reconstructed full-face and difference statistics. They reuse the original retroreflecting environments.

Rule	Quantity	Degree	Intercept	p_{fit}
S	r_N	1	0.668901 ± 0.001808	0.68
S	r_N	2	0.670848 ± 0.004054	0.488
S	r_D	1	0.973271 ± 0.002694	0.0467
S	r_D	2	0.971685 ± 0.005835	0.014
S	cov_CF_over_N	1	-0.076087 ± 0.000811	0.0462
S	cov_CF_over_N	2	-0.075181 ± 0.001785	0.0158
S	mean_imbalance	1	-0.000008 ± 0.000203	0.822
S	mean_imbalance	2	0.000083 ± 0.000455	0.559
O	r_N	1	0.667421 ± 0.001668	0.0378
O	r_N	2	0.666500 ± 0.003793	0.0109
O	r_D	1	0.968273 ± 0.002474	0.286
O	r_D	2	0.963771 ± 0.005622	0.194
O	cov_CF_over_N	1	-0.075202 ± 0.000749	0.0361
O	cov_CF_over_N	2	-0.074333 ± 0.001678	0.012
O	mean_imbalance	1	0.000074 ± 0.000230	0.926
O	mean_imbalance	2	0.000230 ± 0.000504	0.851

The reconstructed full-face count in the retroreflecting environments uses $N = C + f_L$ and no additional sample. The partial-port comparison is retrospective because it followed the original rejection of 2/3. It is not counted as an additional independent confirmation of the stripe law.

6 Exploratory geometries, third cumulants and spectrum

The diagnostic collection consists of 128,000 calibration environments and 68,000 subsequently held-out environments, for a total of 196,000. The latter were new when the fixed local-response prediction was tested; all later combined and source-matching uses are retrospective. Both rules and all sizes $16, 32, 64, 128, 256$ are retained. The quarter-strip and square absolute ratios are about 0.5% high. Their difference is closer to the geometric target. Neither fact establishes an asymptotic conclusion for $p = 1/4$.

The fresh half-open and independent diagnostic single-stripe third-cumulant estimates lie on opposite sides of the saddle value. The samples cover different sizes; their linear discrepancy is about two combined standard errors. Masks share environments and a full-face third-cumulant contribution, so agreement among masks is correlated evidence.

Table 11: All auxiliary third-cumulant fits for the fresh half-open data, with refined deterministic targets.

Rule	Quantity	Degree	Target	Intercept	p_{fit}
S	h1	1	0.499442417	0.472570 ± 0.016065	0.357
S	h1	2	0.499442417	0.505052 ± 0.048637	0.209

Rule	Quantity	Degree	Target	Intercept	p_{fit}
S	h2	1	0.532821444	0.509247 ± 0.016054	0.857
S	h2	2	0.532821444	0.483424 ± 0.048867	0.921
S	h4	1	0.549715205	0.525261 ± 0.016054	0.728
S	h4	2	0.549715205	0.510619 ± 0.049387	0.463
S	hN	1	0.266666667	0.189273 ± 0.044674	0.691
S	hN	2	0.266666667	0.211106 ± 0.136632	0.399
O	h1	1	0.499442417	0.474267 ± 0.013113	0.938
O	h1	2	0.499442417	0.463725 ± 0.039239	0.83
O	h2	1	0.532821444	0.525337 ± 0.013268	0.692
O	h2	2	0.532821444	0.541223 ± 0.040942	0.451
O	h4	1	0.549715205	0.544327 ± 0.013926	0.866
O	h4	2	0.549715205	0.549277 ± 0.041379	0.601
O	hN	1	0.266666667	0.237602 ± 0.036399	0.874
O	hN	2	0.266666667	0.191800 ± 0.111056	0.781
J	h1	1	0.499442417	0.473576 ± 0.010200	0.822
J	h1	2	0.499442417	0.479796 ± 0.030581	0.562
J	h2	1	0.532821444	0.518688 ± 0.010289	0.899
J	h2	2	0.532821444	0.517985 ± 0.031434	0.711
J	h4	1	0.549715205	0.536241 ± 0.010581	0.888
J	h4	2	0.549715205	0.533323 ± 0.031591	0.762
J	hN	1	0.266666667	0.218150 ± 0.027801	0.89
J	hN	2	0.266666667	0.199247 ± 0.085880	0.849

Table 12: All exploratory diagnostic fits, including both batches separately and together, both rules and descriptive joint fits. No additional confirmation decision is made.

Data	Rule	Observable	Degree	Intercept	p_{fit}
Calibration	S	v_N	1	0.673969 ± 0.006996	0.619
Calibration	S	v_N	2	0.666737 ± 0.012266	0.53
Calibration	S	h_N	1	0.421095 ± 0.099658	0.662
Calibration	S	h_N	2	0.206083 ± 0.223955	0.804
Calibration	S	$v_{1/2,1}$	1	0.803660 ± 0.004009	0.666
Calibration	S	$v_{1/2,1}$	2	0.796983 ± 0.007006	0.895
Calibration	S	Quarter strip	1	0.879956 ± 0.002807	0.217
Calibration	S	Quarter strip	2	0.871540 ± 0.004885	0.992
Calibration	S	Square	1	0.875283 ± 0.002744	0.424
Calibration	S	Square	2	0.869409 ± 0.004774	0.765
Calibration	S	$h_{1/2,1}$	1	0.585006 ± 0.036161	0.748
Calibration	S	$h_{1/2,1}$	2	0.577605 ± 0.080579	0.546
Calibration	S	Square minus strip	1	-0.004634 ± 0.001305	0.266
Calibration	S	Square minus strip	2	-0.002137 ± 0.002272	0.341
Calibration	O	v_N	1	0.661504 ± 0.006960	0.454
Calibration	O	v_N	2	0.663135 ± 0.012339	0.273
Calibration	O	h_N	1	0.185978 ± 0.078446	0.101
Calibration	O	h_N	2	-0.180217 ± 0.184878	0.483
Calibration	O	$v_{1/2,1}$	1	0.799435 ± 0.003975	0.449
Calibration	O	$v_{1/2,1}$	2	0.804737 ± 0.007010	0.406
Calibration	O	Quarter strip	1	0.877267 ± 0.002794	0.217
Calibration	O	Quarter strip	2	0.882834 ± 0.004903	0.282
Calibration	O	Square	1	0.874590 ± 0.002735	0.562
Calibration	O	Square	2	0.878548 ± 0.004797	0.594
Calibration	O	$h_{1/2,1}$	1	0.464553 ± 0.028497	0.697
Calibration	O	$h_{1/2,1}$	2	0.436471 ± 0.064514	0.549
Calibration	O	Square minus strip	1	-0.002676 ± 0.001315	0.5
Calibration	O	Square minus strip	2	-0.004226 ± 0.002291	0.432
Calibration	J	v_N	1	0.667705 ± 0.004934	0.54
Calibration	J	v_N	2	0.664947 ± 0.008699	0.563
Calibration	J	h_N	1	0.275925 ± 0.061641	0.128
Calibration	J	h_N	2	-0.023656 ± 0.142574	0.599
Calibration	J	$v_{1/2,1}$	1	0.801530 ± 0.002822	0.687
Calibration	J	$v_{1/2,1}$	2	0.800858 ± 0.004955	0.756

Data	Rule	Observable	Degree	Intercept	p_{fit}
Calibration	J	Quarter strip	1	0.878606 ± 0.001980	0.228
Calibration	J	Quarter strip	2	0.877166 ± 0.003460	0.39
Calibration	J	Square	1	0.874935 ± 0.001937	0.675
Calibration	J	Square	2	0.873957 ± 0.003384	0.638
Calibration	J	$h_{1/2,1}$	1	0.510701 ± 0.022382	0.219
Calibration	J	$h_{1/2,1}$	2	0.491601 ± 0.050362	0.51
Calibration	J	Square minus strip	1	-0.003663 ± 0.000926	0.385
Calibration	J	Square minus strip	2	-0.003173 ± 0.001613	0.514
Held-out	S	v_N	1	0.661996 ± 0.008225	0.349
Held-out	S	v_N	2	0.664418 ± 0.013435	0.198
Held-out	S	h_N	1	0.220628 ± 0.124529	0.133
Held-out	S	h_N	2	0.367875 ± 0.253102	0.0763
Held-out	S	$v_{1/2,1}$	1	0.804928 ± 0.004750	0.438
Held-out	S	$v_{1/2,1}$	2	0.813086 ± 0.007785	0.618
Held-out	S	Quarter strip	1	0.880615 ± 0.003370	0.52
Held-out	S	Quarter strip	2	0.887197 ± 0.005539	0.991
Held-out	S	Square	1	0.879844 ± 0.003304	0.421
Held-out	S	Square	2	0.886469 ± 0.005448	0.787
Held-out	S	$h_{1/2,1}$	1	0.476540 ± 0.044044	0.867
Held-out	S	$h_{1/2,1}$	2	0.496274 ± 0.090781	0.717
Held-out	S	Square minus strip	1	-0.000774 ± 0.001570	0.584
Held-out	S	Square minus strip	2	-0.000889 ± 0.002559	0.378
Held-out	O	v_N	1	0.663257 ± 0.008260	0.249
Held-out	O	v_N	2	0.676727 ± 0.013521	0.282
Held-out	O	h_N	1	0.287197 ± 0.105017	0.598
Held-out	O	h_N	2	0.264273 ± 0.211424	0.394
Held-out	O	$v_{1/2,1}$	1	0.800744 ± 0.004751	0.352
Held-out	O	$v_{1/2,1}$	2	0.802077 ± 0.007785	0.2
Held-out	O	Quarter strip	1	0.878155 ± 0.003368	0.4
Held-out	O	Quarter strip	2	0.875162 ± 0.005525	0.289
Held-out	O	Square	1	0.875503 ± 0.003279	0.249
Held-out	O	Square	2	0.871111 ± 0.005392	0.216
Held-out	O	$h_{1/2,1}$	1	0.544151 ± 0.037747	0.732
Held-out	O	$h_{1/2,1}$	2	0.533293 ± 0.075626	0.532
Held-out	O	Square minus strip	1	-0.002674 ± 0.001570	0.885
Held-out	O	Square minus strip	2	-0.004047 ± 0.002558	0.91
Held-out	J	v_N	1	0.662624 ± 0.005828	0.387
Held-out	J	v_N	2	0.670534 ± 0.009530	0.288
Held-out	J	h_N	1	0.259531 ± 0.080281	0.366
Held-out	J	h_N	2	0.306852 ± 0.162261	0.213
Held-out	J	$v_{1/2,1}$	1	0.802837 ± 0.003359	0.498
Held-out	J	$v_{1/2,1}$	2	0.807581 ± 0.005505	0.394
Held-out	J	Quarter strip	1	0.879385 ± 0.002382	0.602
Held-out	J	Quarter strip	2	0.881165 ± 0.003912	0.432
Held-out	J	Square	1	0.877657 ± 0.002328	0.35
Held-out	J	Square	2	0.878710 ± 0.003833	0.182
Held-out	J	$h_{1/2,1}$	1	0.515520 ± 0.028661	0.848
Held-out	J	$h_{1/2,1}$	2	0.518127 ± 0.058105	0.846
Held-out	J	Square minus strip	1	-0.001724 ± 0.001110	0.853
Held-out	J	Square minus strip	2	-0.002469 ± 0.001809	0.716
Combined	S	v_N	1	0.668692 ± 0.005310	0.706
Combined	S	v_N	2	0.664796 ± 0.009041	0.572
Combined	S	h_N	1	0.339106 ± 0.077755	0.265
Combined	S	h_N	2	0.250579 ± 0.167732	0.164
Combined	S	$v_{1/2,1}$	1	0.804012 ± 0.003053	0.968
Combined	S	$v_{1/2,1}$	2	0.803617 ± 0.005197	0.883
Combined	S	Quarter strip	1	0.880268 ± 0.002149	0.883
Combined	S	Quarter strip	2	0.878029 ± 0.003656	0.958
Combined	S	Square	1	0.877076 ± 0.002104	0.929
Combined	S	Square	2	0.876735 ± 0.003585	0.802
Combined	S	$h_{1/2,1}$	1	0.543120 ± 0.027936	0.893
Combined	S	$h_{1/2,1}$	2	0.537200 ± 0.060256	0.74
Combined	S	Square minus strip	1	-0.003181 ± 0.001000	0.158
Combined	S	Square minus strip	2	-0.001381 ± 0.001695	0.176

Data	Rule	Observable	Degree	Intercept	p_{fit}
Combined	O	v_N	1	0.662343 ± 0.005308	0.117
Combined	O	v_N	2	0.668798 ± 0.009093	0.0773
Combined	O	h_N	1	0.223005 ± 0.063110	0.485
Combined	O	h_N	2	0.031525 ± 0.139835	0.955
Combined	O	$v_{1/2,1}$	1	0.800010 ± 0.003040	0.14
Combined	O	$v_{1/2,1}$	2	0.803666 ± 0.005194	0.0941
Combined	O	Quarter strip	1	0.877589 ± 0.002144	0.147
Combined	O	Quarter strip	2	0.879681 ± 0.003657	0.0879
Combined	O	Square	1	0.874959 ± 0.002094	0.303
Combined	O	Square	2	0.875560 ± 0.003573	0.166
Combined	O	$h_{1/2,1}$	1	0.496337 ± 0.022771	0.767
Combined	O	$h_{1/2,1}$	2	0.485272 ± 0.049088	0.583
Combined	O	Square minus strip	1	-0.002661 ± 0.001005	0.637
Combined	O	Square minus strip	2	-0.004122 ± 0.001702	0.753
Combined	J	v_N	1	0.665516 ± 0.003754	0.333
Combined	J	v_N	2	0.666786 ± 0.006411	0.275
Combined	J	h_N	1	0.269114 ± 0.049001	0.354
Combined	J	h_N	2	0.121345 ± 0.107406	0.452
Combined	J	$v_{1/2,1}$	1	0.802002 ± 0.002154	0.472
Combined	J	$v_{1/2,1}$	2	0.803641 ± 0.003674	0.419
Combined	J	Quarter strip	1	0.878925 ± 0.001518	0.45
Combined	J	Quarter strip	2	0.878854 ± 0.002585	0.409
Combined	J	Square	1	0.876012 ± 0.001484	0.708
Combined	J	Square	2	0.876146 ± 0.002530	0.537
Combined	J	$h_{1/2,1}$	1	0.515012 ± 0.017650	0.841
Combined	J	$h_{1/2,1}$	2	0.505987 ± 0.038058	0.831
Combined	J	Square minus strip	1	-0.002922 ± 0.000709	0.425
Combined	J	Square minus strip	2	-0.002746 ± 0.001201	0.376

Table 13: All auxiliary spatial-spectrum fits. Modes share environments and are correlated.

Rule	Mode	Degree	Intercept	p_{fit}
S	1	1	0.923333 ± 0.001941	0.549
S	1	2	0.925177 ± 0.004157	0.331
S	2	1	0.961187 ± 0.002019	0.229
S	2	2	0.955687 ± 0.004325	0.35
S	4	1	0.983274 ± 0.002046	0.0857
S	4	2	0.975058 ± 0.004382	0.519
S	8	1	0.995469 ± 0.002060	0.599
S	8	2	0.993938 ± 0.004433	0.35
O	1	1	0.920420 ± 0.001944	0.306
O	1	2	0.921088 ± 0.004148	0.126
O	2	1	0.963107 ± 0.002020	0.127
O	2	2	0.965689 ± 0.004328	0.0554
O	4	1	0.982540 ± 0.002062	0.388
O	4	2	0.978145 ± 0.004427	0.426
O	8	1	0.990160 ± 0.002067	0.317
O	8	2	0.985719 ± 0.004421	0.316
J	1	1	0.921879 ± 0.001374	0.455
J	1	2	0.923128 ± 0.002936	0.288
J	2	1	0.962147 ± 0.001428	0.185
J	2	2	0.960684 ± 0.003059	0.0653
J	4	1	0.982910 ± 0.001452	0.231
J	4	2	0.976585 ± 0.003114	0.73
J	8	1	0.992824 ± 0.001459	0.25
J	8	2	0.989818 ± 0.003130	0.308

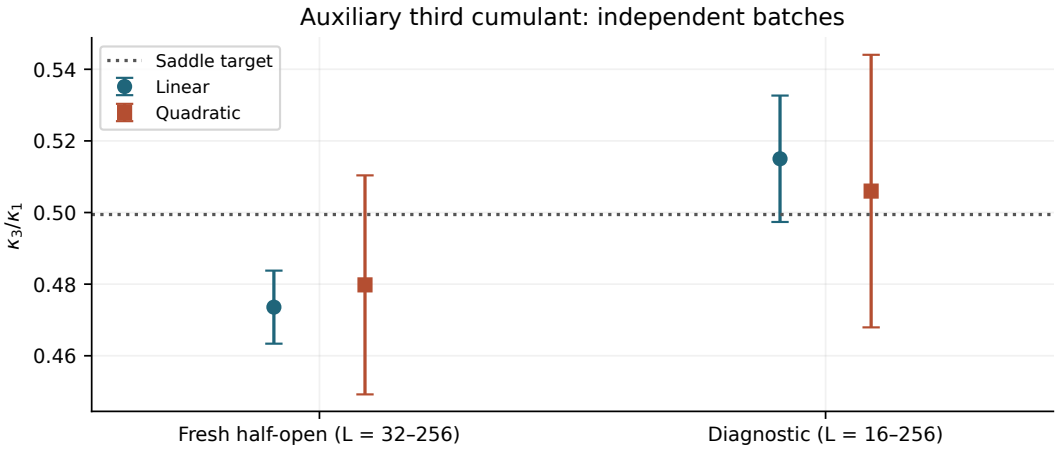


Figure 6: Auxiliary third-cumulant estimates from two independent collections, with both extrapolations and one-standard-error bars.

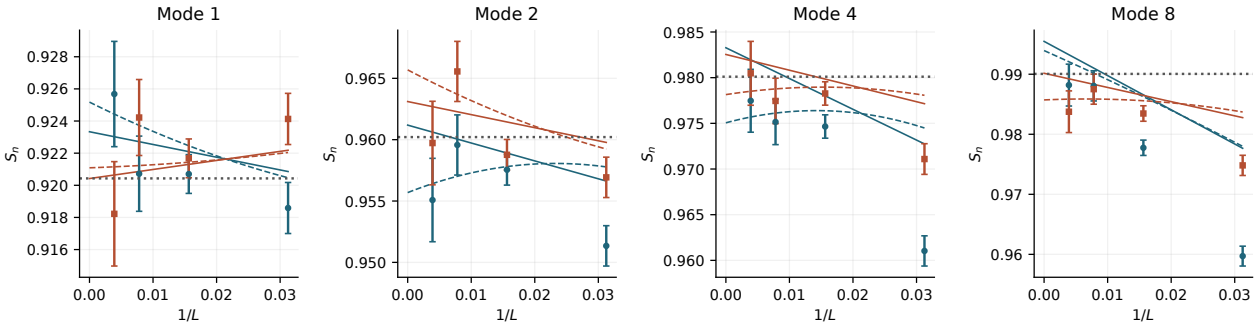


Figure 7: Auxiliary transverse spectrum, all sizes and both rules. The continuum target is $S_n = 1 - I(2\pi n)$. These correlated modes do not enter the primary confirmation family.

7 Why the finite-size program stopped

Specified finite-range fourth-order response moments were calibrated on the first diagnostic sample. The fixed range-two prediction was then tested on the held-out sample. The four joint probabilities were approximately $0.303, 0.288, 0.620, 0.612$, with no Holm rejection. This means that these tests did not exclude that candidate response model at their attained resolution. Coefficients depended on the chosen subtraction and range; they were not unique microscopic couplings. At $L = 256$, changing range from one to three changed one quarter-strip prediction by about 0.000674 ± 0.000164, larger than the desired 10^{-4} model-error scale.

Subsequent work measured all output separations, separated near and far contributions with their covariance, and checked the exact cancellation of artificial range dependence in their sum. Reconstructing a measured window variance from the same complete measured response is an identity, not an independent finite-parameter prediction. Fourth- and sixth-order source matching, selected bulk, surface and contact-line terms, and specified two-loop calculations were pursued. The resulting seven-parameter subset was full rank, with a condition number of order 10^3. It must not be described as algebraically rank deficient. That does not establish complete physical identifiability or control of all operators of the same order.

The resulting 264 candidate finite-size predictions had statistical covariances but no bounded physical remainder. The corrected single-stripe third-cumulant linear/quadratic intercepts, approximately 0.501981 and 0.499560, did not explain the older downward estimate near 0.474. Numerical mesh refinement of a continuum calculation is not physical size refinement of the microscopic model. The proposed subsequent confirmation was therefore never frozen or sampled. The user stopped further research and chose a public report of the limited results. No correction from this unfinished program is applied to the main estimates or their uncertainties.

8 Reproduction and evidential limits

The repository provides a numeric source registry with JSON pointers, file hashes, exact original seed allocations, a standalone sampler and the original statistical cores with portable input adapters. The verification script rebuilds all tables and figures, checks numeric sources and repeats the closed-form calculation. In a new Python 3.12 environment with the locked dependencies, replay of both stripe bootstraps, both extrapolations and all associated decisions gave zero difference from the stored outputs in 349,312 numeric comparisons. Four original blocks, containing 4,500 environments across both rules and both stripe samplers, were regenerated and compared element by element; all were identical. These are replays of existing seeds, not new samples.

This release does not replay all original-geometry raw data, every auxiliary spectrum or higher-source statistic, or the unfinished finite-size program. Their saved results are identified as such. Other operating systems and other numerical-library versions have not been tested. The published hashes do not independently date the local registration. Separate implementations and exact identities reduce implementation risk; they do not replace external expert review or a proof of a microscopic continuum limit.

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